

Code-Layout, Readability and Reusability

Date:	30 June 2026
Team ID:	XXXXXX
Project Name:	Voice-Based Concept Understanding Analyser
Maximum Marks:	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Organized folders for frontend, backend, models, utilities, and reports	Yes	PEP-8 followed
2	Proper File Structure	Organized folders for frontend, backend, models, utilities, and reports	Yes	Modular project structure
3	Meaningful Variable Names	Descriptive and self-explanatory variable names	Yes	Easy to understand and maintain

4	Function / Method Names	Clear and descriptive function names	Yes	Follows Python naming conventions
5	Code Comments	Comments added for complex logic and important functions	Yes	Well documented
6	Modular Design	Independent reusable modules for transcription, semantic analysis, audio processing, and reporting	Yes	High code reusability
7	No Redundant Code	Common logic reused through utility functions	Yes	Duplicate code minimized
8	Error Handling	Proper exception handling for file uploads, audio processing, and model execution	Yes	Robust validation implemented

Reusable Components / Modules

S.No	Component	Technology	Reusability
1	User Authentication Module	Python, Streamlit	High

2	Audio Upload & Recording Module	Streamlit	High
3	Speech-to-Text Module	OpenAI Whisper	High
4	Semantic Similarity Module	Sentence-BERT API	High
5	Audio Feature Extraction Module	Librosa	High
6	Evaluation & Scoring Engine	Python	High